EE-482: Network Security and Cryptography

LECTURES NO 6: *Data Encryption Standard (DES)*

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Organization of Lecture

1. Data Encryption Standard (DES)
2. Simplified Data Encryption Standard (S-DES)
3. Key Generation of S-DES
4. S-DES Encryption Algorithm
5. S-DES Decryption Algorithm

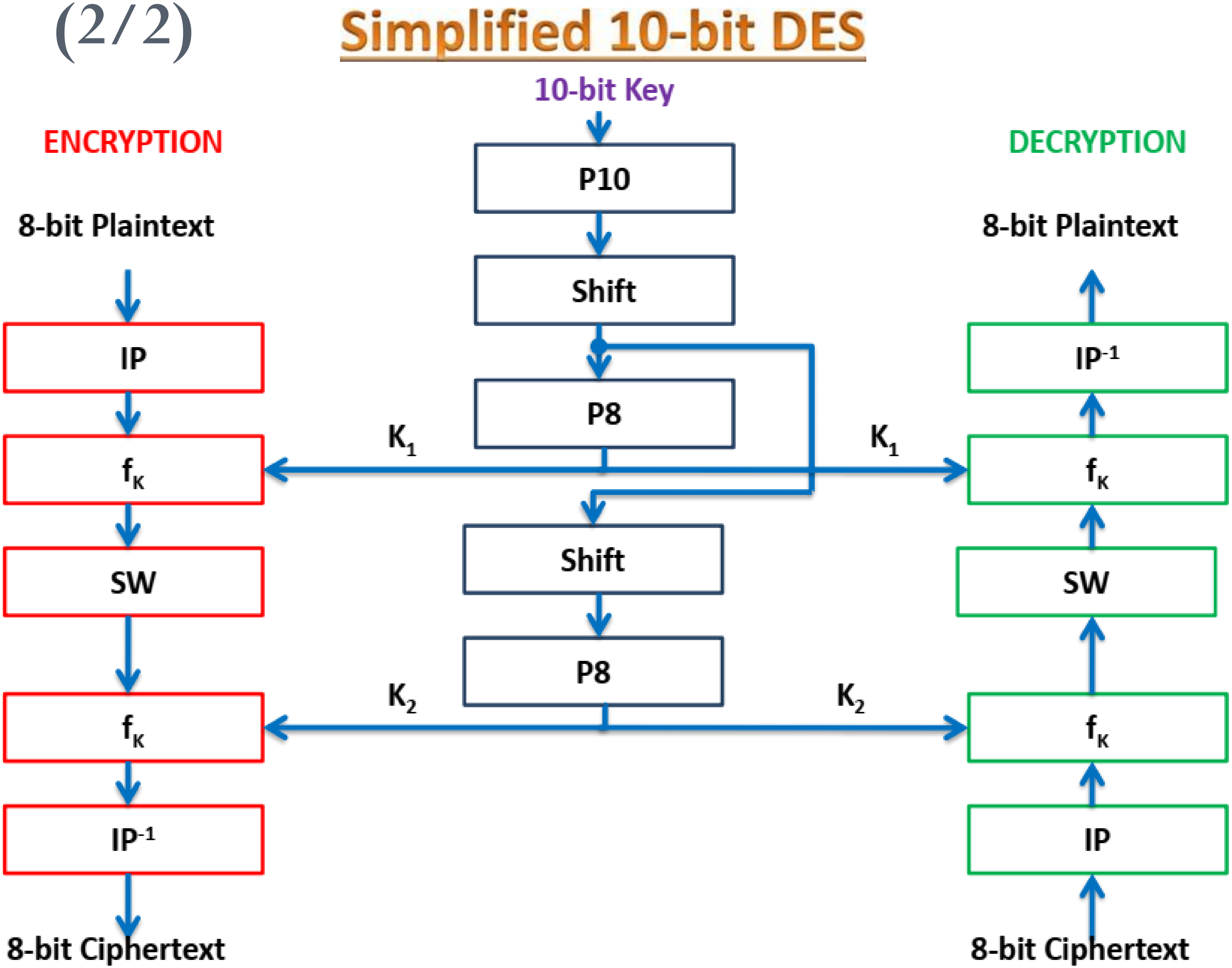
1. Data Encryption Standard (DES)

- The most widely used encryption scheme.
- DES is also sometimes called as ***Data Encryption Algorithm (DEA)***.
- It was initially used by U.S government applications.
- Used as a cryptographic algorithm for over 2 decades but now DES has been found vulnerable against powerful attacks.
- DES is a block cipher.
- The plaintext is processed in 64-bit blocks.
- The key is 56-bits in length.

2. Simplified Data Encryption Standard (S-DES) (1/2)

- Uses 8-bit plaintext block.
- Uses two keys (each of 8 bit) i.e., K_1 and K_2.
- Output is 8-bit cipher text.
- Some important abbreviations used in S-DES Encryption and decryption algorithms are:
  - *IP = Initial Permutation*
  - *E/P = Extended Permutation*
  - *B E/P = Before Extended Permutation*
  - *S-Matrices/S-boxes = Substitution Matrices/Substitution Boxes*
  - S_0 = *Substitution Matrix* S_0
  - S_1 = *Substitution Matrix* S_1
  - *P4 = Permutation 4*
  - IP^{-1} = *Inverse Initial Permutation*
  - f_k = *Complex Function*

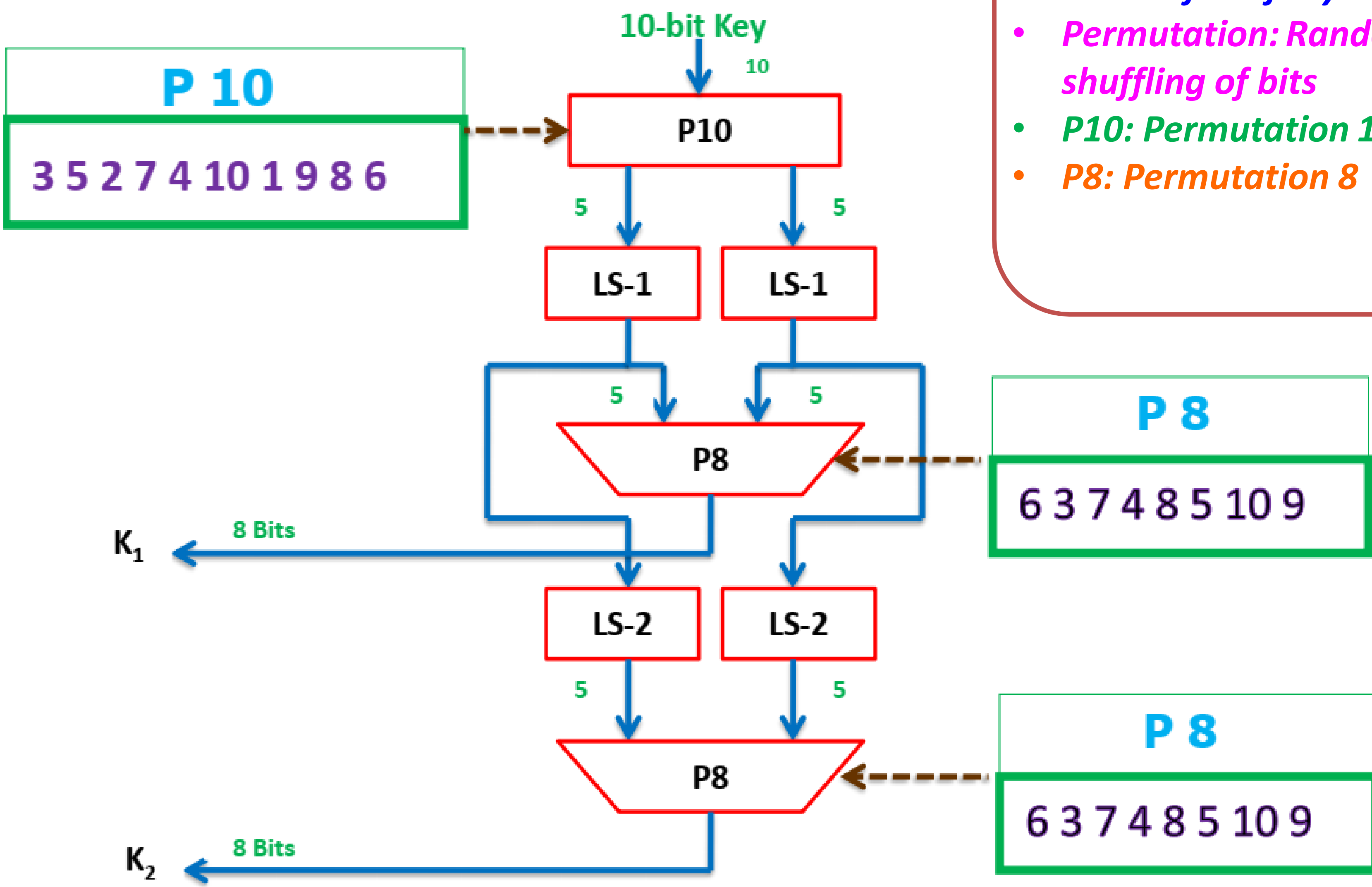
2. Simplified Data Encryption Standard (S-DES) (2/2)



3. Key Generation of S-DES (1/4)

- *LS-1: Left shift by 1-Bit*
- *LS-2: Left shift by 2-Bits*
- *Permutation: Random shuffling of bits*
- *P10: Permutation 10*
- *P8: Permutation 8*

Key Generation



3. Key Generation of S-DES (2/4)

Example1: * Let the Plain text be:

1st digit	2nd digit	3rd	4th	5th	6th	7th	8th	9th	10th
1	0	1	0	0	0	0	0	1	0

* Let the 10th Permutation be:

P10: 3 5 2 7 4 10 1 9 8 6

* Let the 8th Permutation be:

P8: 6 3 7 4 8 5 10 9

Generate Keys K_1 and K_2.

Solution

① Compute P10:

No.s	3	5	2	7	4	10	1	9	8	6
P10	1	0	0	0	0	0	1	1	0	0

3. Key Generation of S-DES (3/4)

② Divide P10 computed above into two groups of 5 bits each

G1

1	2	3	4	5
1	0	0	0	0

G2

6	7	8	9	10
0	1	1	0	0

③ Left shift by 1-bit to G1 & G2

1	2	3	4	5
0	0	0	0	1

G3

6	7	8	9	10
1	1	0	0	0

G4

3. Key Generation of S-DES (4/4)

④ Apply P8 on step 3

No.s	6	3	7	4	8	5	10	9
P8	1	0	1	0	0	1	0	0

So, Key, K_1 = 1 0 1 0 0 1 0 0

⑤ Left shift by 2-bits to G3 & G4 obtained in step3.

1	2	3	4	5
0	0	1	0	0

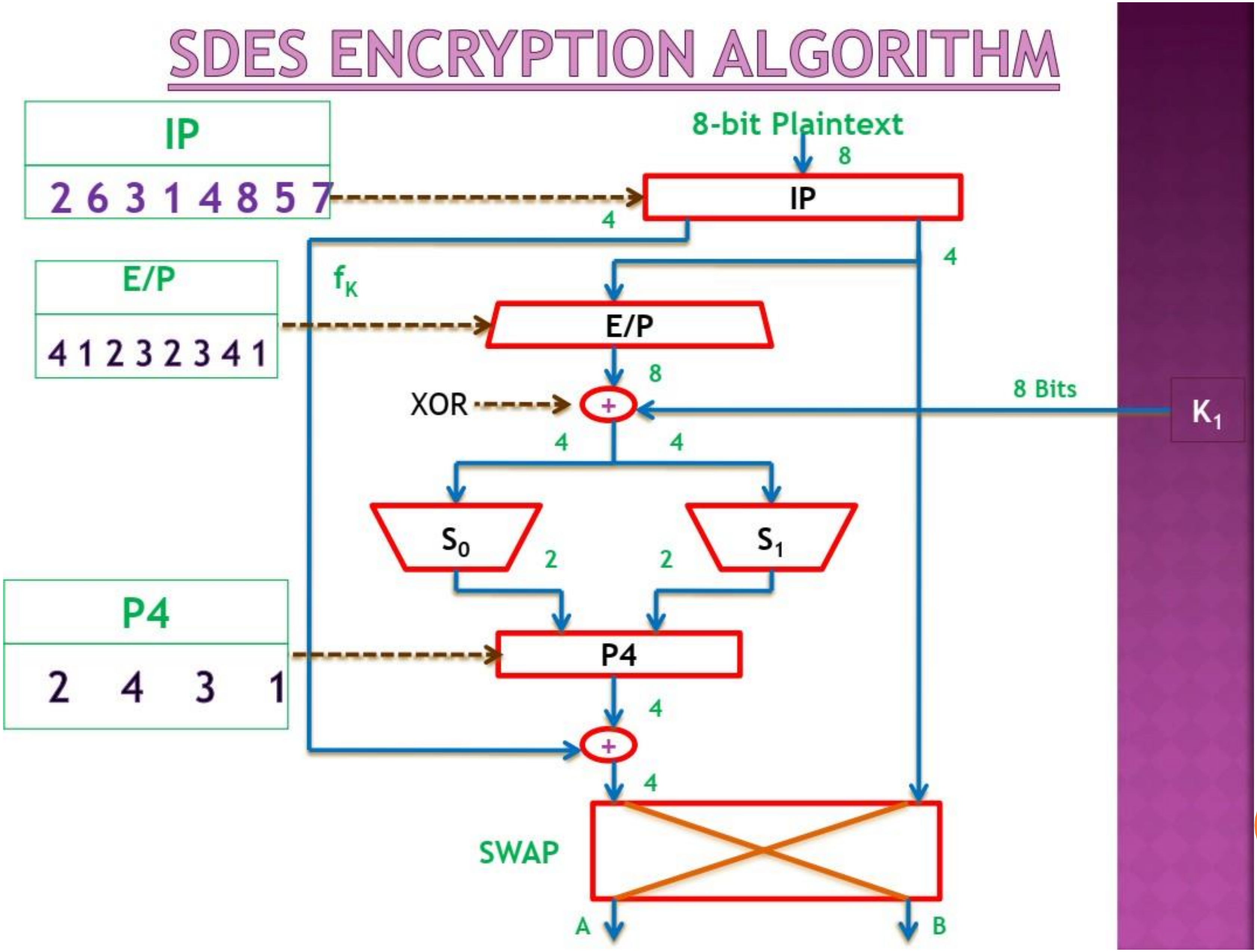
6	7	8	9	10
0	0	0	1	1

⑥ Apply P8 on step 5

No.s	6	3	7	4	8	5	10	9
P8	0	1	0	0	0	0	1	1

So, key, K_2 = 0 1 0 0 0 0 1 1

4. S-DES Encryption Algorithm (1/14)



4. S-DES Encryption Algorithm (2/14)

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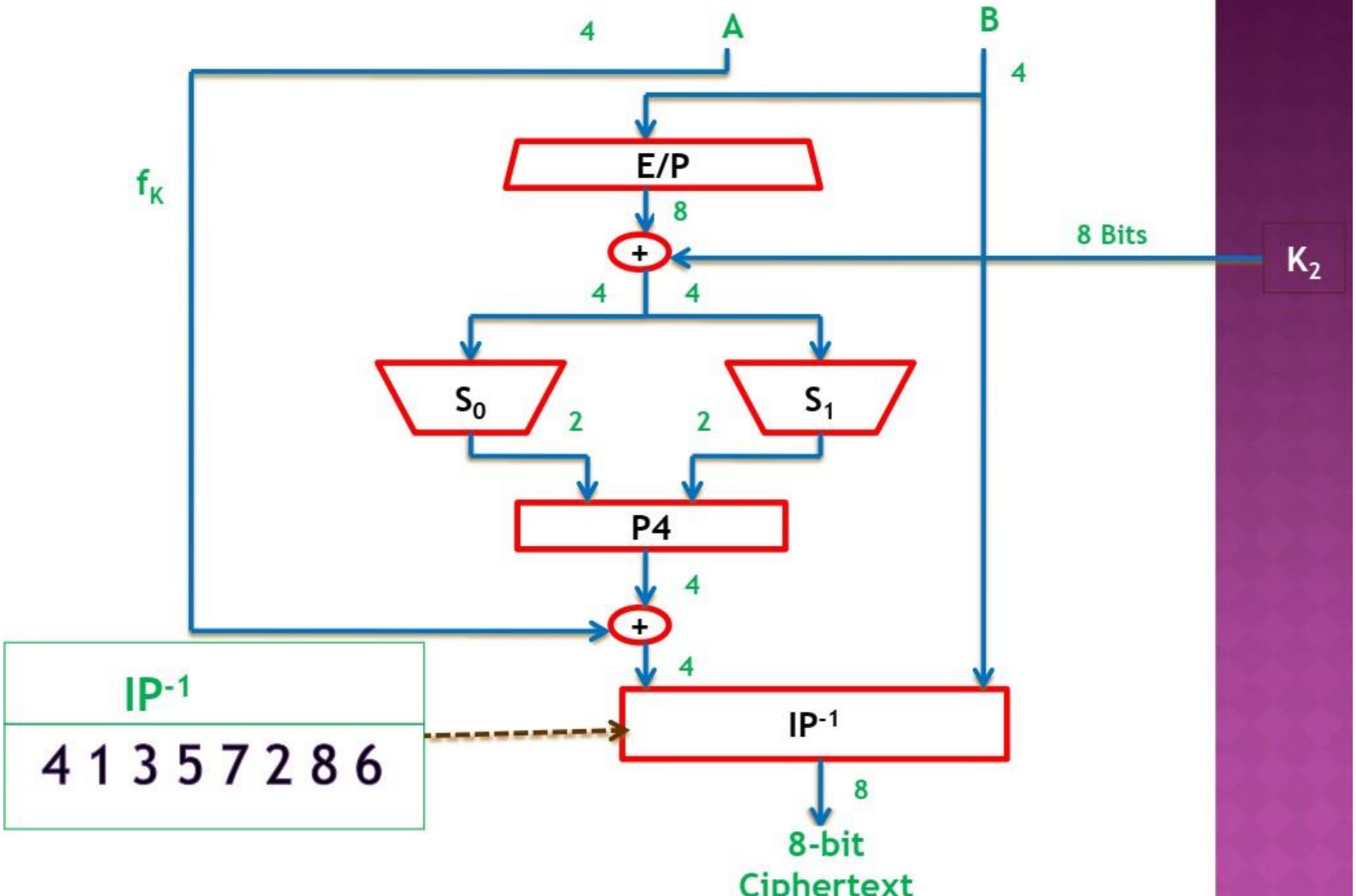


4. S-DES Encryption Algorithm (3/14)

Example 2:

$$\begin{array}{cccccccc} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \end{array}$$

* Let the plaintext be: 0 1 1 0 0 1 1 0
* " " Key 1 (K_1) be: 1 0 1 0 0 1 0 0
* " " Key 2 (K_2) be: 0 1 0 0 0 0 1 1
* Let the IP be: 2 6 3 1 4 8 5 7
* " " E/P be: 4 1 2 3 2 3 4 1
* Let the IP^{-1} be: 4 1 3 5 7 2 8 6
* Let the P4 be: 2 4 3 1
* Substitution Matrices are:

$$S_0 = \begin{bmatrix} 1 & 0 & 3 & 2 \\ 3 & 2 & 1 & 0 \\ 0 & 2 & 1 & 3 \\ 0 & 1 & 3 & 2 \end{bmatrix} \quad \& \quad S_1 = \begin{bmatrix} 0 & 1 & 2 & 3 \\ 2 & 0 & 1 & 3 \\ 3 & 0 & 1 & 0 \\ 2 & 1 & 0 & 3 \end{bmatrix}$$

What will be the secured text for transmission?

4. S-DES Encryption Algorithm (4/14)

Solution

① Apply IP on plain text

No.s	2	6	3	1	4	8	5	7
IP	1	1	1	0	0	0	0	1

② Divide/Split IP in step 1 in two groups of 4-bits each (one group is f_K & other is B E/P)

No.s	1	2	3	4
f_K	1	1	1	0

$\underbrace{\hphantom{xxxxxx}}_{G1}$

No.s	1	2	3	4
B E/P	0	0	0	1

$\underbrace{\hphantom{xxxxxx}}_{G2}$

4. S-DES Encryption Algorithm (5/14)

③ Apply E/P on G2 obtained in step2.

No.s	4	1	2	3	2	3	4	1
E/P	1	0	0	0	0	0	1	0

④ Now perform "XOR" operation b/w E/P found in step3 and key "K_1"

E/P	1	0	0	0	0	0	1	0
K_1	1	0	1	0	0	1	0	0
$(E/P \oplus K_1)$	0	0	1	0	0	1	1	0

⑤ Now split $(E/P \oplus K_1)$ found in step 4 into two groups of each 4-bits

No.s	1	2	3	4
S_0	0	0	1	0

columns (01)

rows (00)

No.s	1	2	3	4
S_1	0	1	1	0

columns (11)

rows (00)

4. S-DES Encryption Algorithm (6/14)

(6) Apply S_0 & S_1 substitution matrices on S_0 & S_1 results obtained in step 5.

$$
S_0 = \begin{array}{c} \text{Rows} \\ 00 \\ 01 \\ 10 \\ 11 \end{array}
\begin{array}{c} \text{Columns} \\ \begin{array}{cccc} 00 & 01 & 10 & 11 \end{array} \\ \begin{bmatrix} 1 & \boxed{0} & 3 & 2 \\ 3 & 2 & 1 & 0 \\ 0 & 2 & 1 & 3 \\ 0 & 1 & 3 & 2 \end{bmatrix} \end{array}
\quad \& \quad
S_1 = \begin{array}{c} \text{Rows} \\ 00 \\ 01 \\ 10 \\ 11 \end{array}
\begin{array}{c} \text{columns} \\ \begin{array}{cccc} 00 & 01 & 10 & 11 \end{array} \\ \begin{bmatrix} 0 & 1 & 2 & \boxed{3} \\ 2 & 0 & 1 & 3 \\ 3 & 0 & 1 & 0 \\ 2 & 1 & 0 & 3 \end{bmatrix} \end{array}
$$

$S_0 = 0$; in binary $S_0 = 00$

$S_1 = 3$; in binary $S_1 = 11$

(7) Combining S_0 & S_1

No.s	1	2	3	4
S_0, S_1	0	0	1	1

4. S-DES Encryption Algorithm (7/14)

(8) Apply P4 on result obtained in step 7

No.s	2	4	3	1
P4	0	1	1	0

(9) Apply XOR operation b/w P4 obtained in step 8 and G1 in step 2.

P4	0	1	1	0
G1	1	1	1	0
P4 ⊕ G1	1	0	0	0

4. S-DES Encryption Algorithm (8/14)

(10) Swap result found in step 9 with group G2 found in step 2.

No.S	1	2	3	4
P4 ⊕ G1	1	0	0	0

G 3

No.S	1	2	3	4
B E/P	0	0	0	1

G 4

No.S	1	2	3	4
B E/P	0	0	0	1

G5

No.S	1	2	3	4
P4 ⊕ G1	1	0	0	0

G6

(11) Apply E/P on G6 found in step 10.

No.S	4	1	2	3	2	3	4	1
E/P	0	1	0	0	0	0	0	1

4. S-DES Encryption Algorithm (9/14)

(12) Now perform "XOR" operation b/w E/P obtained step 11 and Key, K_2 = 0 1 0 0 0 0 1 1

E/P	0	1	0	0	0	0	0	1
K_2	0	1	0	0	0	0	1	1
$(E/P \oplus K_2)$	0	0	0	0	0	0	1	0

(13) Now split $[E/P \oplus K_2]$ found in step 12 into two groups of 4-bits each

No.s	1	2	3	4
S_0	0	0	0	0

column (00)

Row (00)

No.s	1	2	3	4
S_1	0	0	1	0

column (01)

Row (00)

4. S-DES Encryption Algorithm (10/14)

(14) Apply S-matrices (S_0 & S_1) on results obtained in step 13

$$S_0 = \begin{array}{c c} & \text{col}(00) \\ \text{Row}(00) & \begin{bmatrix} \boxed{1} & 0 & 3 & 2 \\ 3 & 2 & 1 & 0 \\ 0 & 2 & 1 & 3 \\ 0 & 1 & 3 & 2 \end{bmatrix} \end{array} \quad \& \quad S_1 = \begin{array}{c c} & \text{col}(01) \\ \text{Row}(00) & \begin{bmatrix} 0 & \boxed{1} & 2 & 3 \\ 2 & 0 & 1 & 3 \\ 3 & 0 & 1 & 0 \\ 2 & 1 & 0 & 3 \end{bmatrix} \end{array}$$

from S-boxes it is clear that:

$S_0 = 1$; in binary, $S_0 = 01$

$S_1 = 1$; in binary, $S_1 = 01$

(15) Combining S_0 & S_1

NO.S	1	2	3	4
S_0, S_1	0	1	0	1

4. S-DES Encryption Algorithm (11/14)

(16) Apply P4 on results obtained in step 15.

No.s	2	4	3	1
P4	1	1	0	0

(17) Apply XOR operation b/w P4 obtained in step 16 and G5 obtained in step 10

P4	1	1	0	0
B E/P or G5	0	0	0	1
$P4 \oplus f_K$	1	1	0	1

4. S-DES Encryption Algorithm (12/14)

(18) Combine $(P_4 \oplus f_k)$ → step 17 with 'G6 found in step 10.

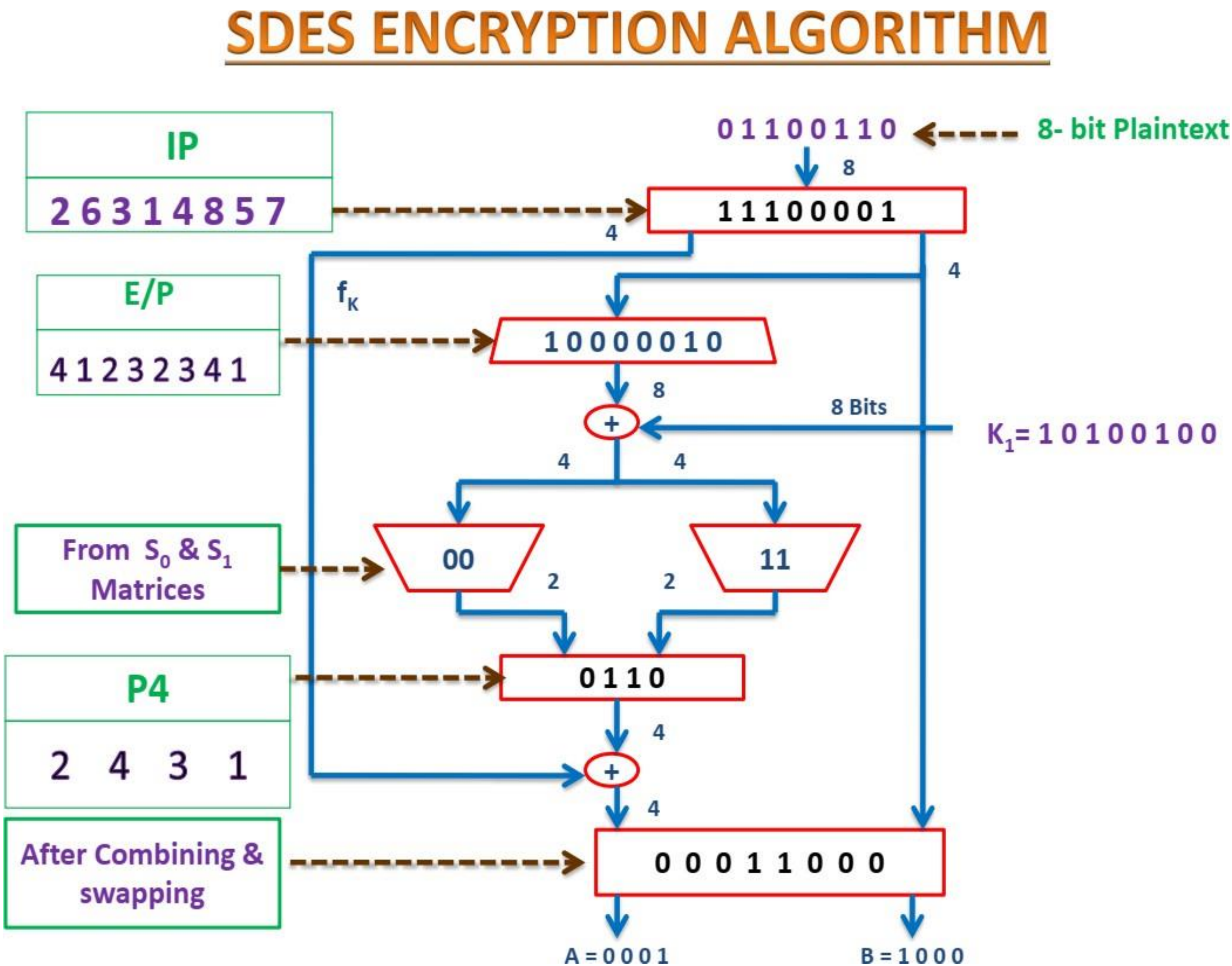
No.s	1	2	3	4	5	6	7	8
	1	1	0	1	1	0	0	0

(19) Now Apply IP^{-1} at step 18 data to get cipher-text

No.s	4	1	3	5	7	2	8	6
IP^{-1}	1	1	0	1	0	1	0	0

Cipher Text: 1 1 0 1 0 1 0 0

4. S-DES Encryption Algorithm (13/14)



4. S-DES Encryption Algorithm (14/14)

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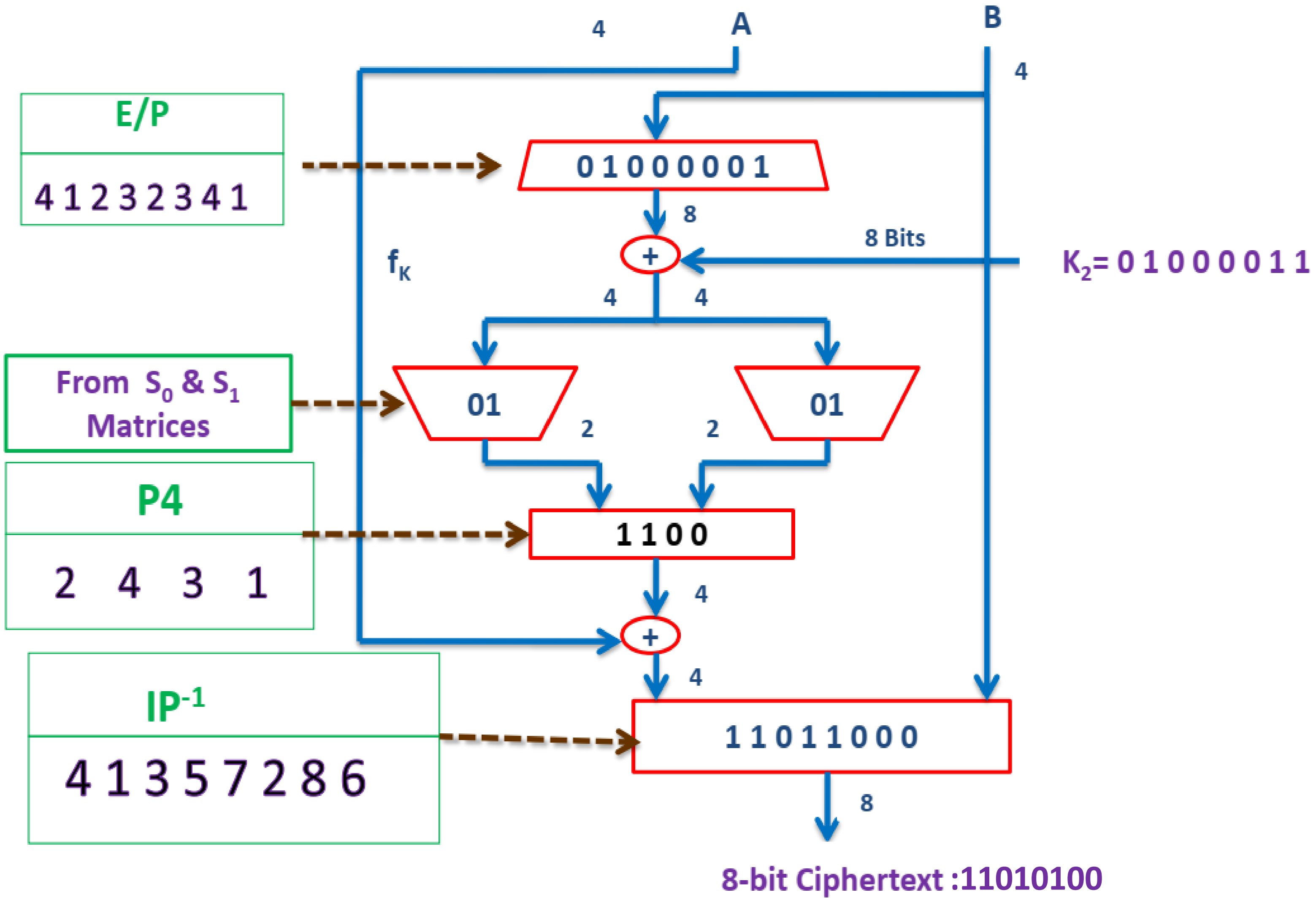

5. S-DES Decryption Algorithm

- Exactly same block diagram as S-DES encryption (Refer to slide 11 and slide 12) with only three exceptions i.e., ciphertext will be used as plaintext, Key 1 will be replaced by Key 2 and Key 2 will be replaced by Key 1.
  - ***Cipher text will be used as a plaintext***
  - $K_1 = K_2$
  - $K_2 = K_1$

References: Textbooks

1. *“Cryptography and Network Security”, by Behrouz A. Forouzan, Debdeep Mukhopadhyay, 2nd Edition, McGraw Hill Education (India) Private Limited.*
2. *“Cryptography and Network Security Principles and Practice”, by William stallings, 6th Edition, Pearson.*
3. *“Computer Security Principles and Practice”, by William stallings, Lawrie Brown, 4th Edition, Pearson.*